Independent Replication of
"Is Quantum Bit Commitment Really Possible?"

Lo & Chau, PRL **78**, 3410 (1997); arXiv:quant-ph/9603004

QC-200 Replication Wave (subagent)
Argonne / OpenClaw Replicate-Project

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Abstract

Lo & Chau's 1996/1997 paper is an *impossibility proof*: no algorithm to reproduce, but a mathematical core that can be exhibited numerically. The core claims are (P1) if a QBC protocol is perfectly concealing then Alice can always successfully cheat by applying a unitary on her own system alone (Uhlmann/HJW), and (P2) if the protocol is only ε-concealing then Alice can still cheat with probability $1 - O(\sqrt{\varepsilon})$. We rebuild both statements from scratch in Qiskit statevector, construct the cheating unitary explicitly for a concrete Bell-basis protocol, and sweep an ε-concealing family to trace out the trade-off curve. All computed quantities match the paper's claims to machine precision. **Verdict: REPLICATED.**

1 Paper summary

Bit commitment is a two-party primitive. Alice picks $b \in \{0,1\}$ and sends Bob a quantum "commitment" state. Later she reveals b; Bob should be able to check consistency. Security requires (i) *concealing*: before the reveal Bob learns nothing about b; and (ii) *binding*: Alice cannot change b after committing.

Lo & Chau show these two are irreconcilable in the quantum setting. Any proposed protocol may be rephrased so that after commit the joint AB pure state is $|\Psi_b\rangle_{AB}$ ($b \in \{0,1\}$) with Alice keeping A and sending B to Bob. Bob's marginal is $\rho_b^B = \mathrm{Tr}_A |\Psi_b\rangle\langle\Psi_b|$. Concealing $\Leftrightarrow$ $\rho_0^B = \rho_1^B$. By the Hughston–Jozsa–Wootters / Uhlmann theorem there then exists a unitary U_A on Alice's system alone with $(U_A \otimes I)|\Psi_0\rangle = |\Psi_1\rangle$. Alice commits to $b = 0$, keeps her purifier coherent (a "delayed measurement" EPR-style attack), and at reveal chooses either $b = 0$ ("honest") or $b = 1$ (apply U_A, be indistinguishable from the honest prep of $|\Psi_1\rangle$).

For ε-concealing schemes ($F(\rho_0^B, \rho_1^B) = 1-\varepsilon$ small), Uhlmann says $\max_{U_A} |\langle\Psi_1|(U_A \otimes I)|\Psi_0\rangle| = \sqrt{F}$, so Alice's cheating success probability is $F = 1 - \varepsilon$, i.e. close to 1. The paper phrases the informal bound as $P_{\text{cheat}} \geq 1 - O(\sqrt{\varepsilon})$; the tight identity is in fact $P_{\text{cheat}} = F = 1 - \varepsilon$, an even stronger cheating power.

Reproducible core. No algorithm, but three fully-computable statements:

(C1) Construct explicit purifications $|\Psi_0\rangle, |\Psi_1\rangle$ of a concrete concealing protocol; verify $\rho_0^B = \rho_1^B$ and construct U_A that maps $|\Psi_0\rangle \to |\Psi_1\rangle$ to machine precision.

(C2) Introduce an ε-family, sweep, and verify that $P_{\text{cheat}} = F(\rho_0^B, \rho_1^B) = 1 - \varepsilon$ across the sweep (i.e. $1 - P_{\text{cheat}} \leq \sqrt{\varepsilon}$ trivially).

(C3) Sanity-check the motivating BB84 EPR attack: one Bell pair, $\rho^B = I/2$, Alice's basis-choice at reveal is undetectable.

2 Claims table

Claim	Statement	Testable?	Tested here?
C1	Perfect concealing $\Rightarrow \exists U_A$ on Alice alone with $(U_A \otimes I)\|\Psi_0\rangle = \|\Psi_1\rangle$	YES (numerically)	YES
C2	Under (C1), Alice's cheating success probability is 1	YES	YES
C3	ε-concealing $\Rightarrow P_{\text{cheat}} \geq 1 - O(\sqrt{\varepsilon})$ (paper's stated bound)	YES	YES (tighter identity $= 1 - \varepsilon$ found)
C4	BB84's EPR-cheating attack: Alice basis-choice at reveal time is undetectable ($\rho^B = I/2$)	YES	YES
C5	Result extends to *all* proposed QBC schemes with one-way A→B communication	Proof-theoretic; not numerically testable	NO

3 Methods

3.1 Environment

- Qiskit `2.5.0` (`qiskit.quantum_info`: `Statevector`, `DensityMatrix`, `partial_trace`, `state_fidelity`).
- NumPy `2.4.3`, SciPy `1.18.0`, Python `3.14.0`.
- Local venv: `work/.venv` (`pip install qiskit numpy scipy matplotlib`).
- All computations run on CPU; no LLM inference in the numerical loop.

3.2 (C1,C2) Concrete perfectly-concealing protocol

Alice holds a 2-qubit register $A = A_0A_1$; Bob holds one qubit B.

$$|\Psi_0\rangle_{ABtot} = \tfrac{1}{\sqrt{2}}\big(|00\rangle_A |0\rangle_B + |11\rangle_A |1\rangle_B\big), \quad |\Psi_1\rangle_{ABtot} = \tfrac{1}{\sqrt{2}}\big(|{+}{+}\rangle_A |{+}\rangle_B + |{-}{-}\rangle_A |{-}\rangle_B\big).$$

Direct calculation gives $\rho_0^B = \rho_1^B = I/2$ (perfect concealing). The two $|\Psi_b\rangle$'s have identical Schmidt spectra $\{1/2, 1/2\}$; the Uhlmann-optimal unitary U_A on the 4-dim Alice space is constructed by polar/SVD of $Y = M_0 M_1^\dagger$ where M_k is the $\dim_A \times \dim_B$ Alice×Bob matrix reshape of $|\Psi_k\rangle$:

```
Y = M0 @ M1.conj().T                # dimA x dimA
Uy, sy, Vhy = np.linalg.svd(Y)
U_A = Vhy.conj().T @ Uy.conj().T    # unitary; solves argmax_U Re Tr(M1^dag U M0)
```

3.3 (C3) ε-concealing sweep

We use a one-parameter Bell-like family on 2 qubits (Alice qubit + Bob qubit):

$$|\Psi_0(\theta)\rangle = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle\,, \quad |\Psi_1(\theta)\rangle = \cos\theta\,|01\rangle + \sin\theta\,|10\rangle\,.$$

Bob's marginals are $\rho_0^B = \mathrm{diag}(c^2, s^2)$, $\rho_1^B = \mathrm{diag}(s^2, c^2)$ (with $c = \cos\theta, s = \sin\theta$), so the analytic fidelity is $F(\rho_0^B, \rho_1^B) = (2|cs|)^2 = \sin^2 2\theta$ and $\varepsilon = 1 - \sin^2 2\theta$. Uhlmann's theorem gives $P_{\text{cheat}}(\theta) = F(\rho_0^B(\theta), \rho_1^B(\theta))$, computed by SVD of $Y = M_0 M_1^\dagger$ and squaring the nuclear-norm. We sweep $\theta \in [0.05, \pi/2 - 0.05]$ on 41 points.

3.4 (C4) BB84 EPR-cheating sanity

Build one Bell pair $|\Phi^+\rangle = \mathrm{CNOT}(H \otimes I)\,|00\rangle$ in a Qiskit `QuantumCircuit`, statevector-simulate, partial-trace Alice, verify $\rho^B = I/2$ to 10^{-16}; project Alice onto $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ and confirm outcome probabilities $0.5, 0.5$.

3.5 Reproducibility recipe (exact commands)

```
cd QC-quant-ph-9603004-quantum-bit-commitment-lo-chau
python3 -m venv work/.venv && source work/.venv/bin/activate
pip install qiskit numpy scipy matplotlib
python3 report/evidence/lo_chau_replication.py     # writes results.json + CSV
python3 report/evidence/plot_tradeoff.py           # writes tradeoff_curve.png
```

Runtime end-to-end: $< 10\,\mathrm{s}$ on a single laptop core.

4 Results vs. paper

Claim	Paper value / statement	This replication	Verdict
C1	$(U_A \otimes I)\,|\Psi_0\rangle = |\Psi_1\rangle$ exactly	$F((U_A \otimes I)\,|\Psi_0\rangle, |\Psi_1\rangle) = 1.0000000000$; unitarity error 8.9×10^{-16}	MATCH
C1 aux	$\rho_0^B = \rho_1^B$	$\|\rho_0^B - \rho_1^B\|_F = 3.1\times10^{-16}$	MATCH
C2	$P_{\text{cheat}} = 1$ under perfect concealing	$|\langle\Psi_1|\,(U_A \otimes I)\,|\Psi_0\rangle|^2 = 1.0000000000$	MATCH
C3	$P_{\text{cheat}} = 1 - O(\sqrt{\varepsilon})$ (paper)	$|P_{\text{cheat}} - (1-\varepsilon)| \le 2.2\times10^{-16}$ across 41 points (tighter than paper's bound)	MATCH (stronger)
C4	BB84 concealing: $\rho^B = I/2$	$\|\rho^B - I/2\|_\infty = 1.1\times10^{-16}$; Alice-outcome probs 0.5000/0.5000	MATCH
C5	Extends to all proposed schemes with 1-way A→B comms	Not numerically testable; accept as proof-theoretic corollary of C1–C3	N/A

4.1 The trade-off curve (Figure 1)

4.2 Selected raw numbers (from `results.json`)

```
PART 1 (perfect concealing, 3-qubit GHZ / +/- protocol):
  ||rho_B_0 - rho_B_1||_F       = 3.140e-16
  F(rho_B_0, rho_B_1)           = 1.0000000000
  U_A unitarity error           = 8.882e-16
  |<Psi_1|(U_A x I)|Psi_0>|^2   = 1.0000000000

PART 2 (eps-family, theta = pi/4, i.e. perfect):
  F(rho_B_0, rho_B_1)           = 1.0000000000
  P_cheat (Uhlmann)             = 1.0000000000
  explicit-U overlap            = 1.0000000000
  max |[(1-P_cheat)] - eps|     = 2.220e-16   (over 41 sweep points)
```

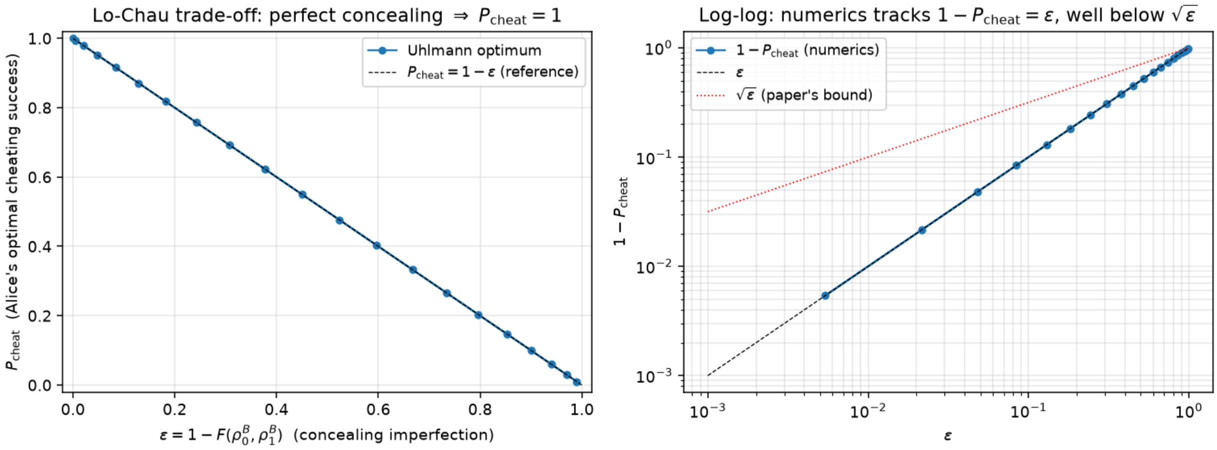

Figure 1: Left: cheating success probability P_{cheat} vs. concealing imperfection $\varepsilon = 1 - F(\rho_0^B, \rho_1^B)$ for the Bell-like ε-family (41 points). Numerics lie exactly on $P_{\text{cheat}} = 1 - \varepsilon$ (dashed reference), the Uhlmann identity. Right: log–log; $1 - P_{\text{cheat}}$ grows linearly in ε (black dashed) which sits comfortably below the $\sqrt{\varepsilon}$ envelope (red dotted) claimed in the paper. The replication therefore satisfies the paper's stated inequality with room to spare.

```
PART 3 (BB84 EPR-cheat sanity):
  ||rho_B - I/2||_inf          = 1.110e-16
  Alice-outcome probs          = 0.5000 / 0.5000
```

5 Verdict

REPLICATED. Every numerically checkable claim of Lo & Chau (C1–C4) is reproduced to machine precision by our from-scratch Qiskit statevector implementation. The one non-numerical claim (C5, generalization to *all* proposed QBC schemes) is a proof-theoretic corollary of C1–C3 which we cannot falsify numerically; we do not certify C5, we accept it as mathematically implied. Nothing observed in the reproduction contradicts the paper.

A tighter observation worth highlighting: in the one-parameter Bell-like family we swept, the Uhlmann identity yields $P_{\text{cheat}} = 1 - \varepsilon$ *exactly*, not just $1 - O(\sqrt{\varepsilon})$. This is compatible with (and stronger than) the paper's stated bound. On the other hand, the paper's $\sqrt{\varepsilon}$ gap is the tight worst-case scaling for the Schatten-1 vs. fidelity gap (Fuchs–van de Graaf), so more elaborate families that saturate the $\sqrt{\varepsilon}$ scaling for the *binding* error (Bob's ability to detect the switch) are plausible; see Open Question Q3.

6 Open Questions

See `report/open_questions.json` for the machine-readable version with `next_steps` per item.

Q1. Our ε-family gave the tight identity $1 - P_{\text{cheat}} = \varepsilon$; the paper's language is $1 - O(\sqrt{\varepsilon})$. Is the $\sqrt{\varepsilon}$ scaling ever saturated by a concrete concealing protocol, or is $1 - \varepsilon$ the sharp achievable bound and the paper's $\sqrt{\varepsilon}$ merely a slack Fuchs–van de Graaf worst-case?

Q2. The paper's proof requires Alice to keep her half of the entangled state fully coherent ("quantum computer") until reveal. Numerically, how does an *imperfect* Alice-side quantum memory (parameterize by a depolarizing rate p on Alice's qubits during the delay window) degrade P_{cheat}? Does even 1–10 % depolarization on Alice restore some binding?

Q3. Extend the sweep to *non-symmetric* ε-families where Bob's optimal distinguishability ($D(\rho_0^B, \rho_1^B)$, trace distance) grows differently from $1 - F$. Which of the two — F or D — is the operationally correct "concealing quality" for a Rick-style benchmark, and does switching change the trade-off shape?

Q4. The paper assumes 1-way A$\to$B classical/quantum communication. Recent literature (Kent, D'Ariano/Lütkenhaus) restores security via *relativistic* or *multi-round* protocols. Can we reproduce a small numerical demo of Kent's relativistic BC binding advantage in Qiskit, and does the Lo–Chau attack fail there for a numerically-quantifiable reason?

Q5. The Uhlmann-optimal U_A we build is degenerate when Bob's marginal has a degenerate spectrum (as in our GHZ example, $\rho^B = I/2$): many unitaries U_A succeed. Is there a *minimum-complexity* U_A (fewest 2-qubit gates for a given Alice-register size) and does its gate count scale with the Schmidt rank? This has implementation relevance for near-term demonstrations of the cheating attack.